

Multi-Touch Marketing Attribution & ROI Dashboard

Project Summary

Project Objective

Developed an interactive marketing analytics dashboard using **Python**, **SQL**, and **Power BI** to analyse customer behaviour, conversion performance, and marketing insights.

Dataset

Used the **E-Commerce Customer Journey: Click to Conversion** dataset containing **12,719 sessions**, **1,872 users**, and **5,050 purchase records** for analysis.

Python

Performed **data cleaning**, **missing value handling**, **duplicate removal**, **data type validation**, and prepared the dataset for analysis using **Pandas**.

SQL

Loaded the cleaned dataset into the database, performed **JOINS**, **Window Functions**, and calculated key business metrics for reporting.

Power BI

Built an interactive dashboard with **KPI Card**, **Customer Conversion Funnel**, **Monthly Purchase Trend**, **Purchases by Country**, **Referral Source Analysis**, **Device Type Distribution**, **Cart-to-Purchase Comparison**, **Slicers**, and **Average Time on Page (97.43 seconds)**.

GitHub

Maintained the project repository with **Jupyter Notebook**, **SQL scripts**, **Power BI dashboard**, and regular commits.

Business Value

Provides insights into customer behaviour, conversion trends, and referral source performance. Helps businesses improve marketing strategies, enhance customer engagement, and increase conversion rates through data-driven decisions.